- A recommender system is a software system that suggests items to users based on their interests, performances, behaviour, or past interactions.
- User → Data/behaviour → recommendation Algo. → Recommended Items
- recommender system is needed because websites/apps have a huge amt. of content. Users cannot manually search through everything.
- They help:
  - Find relevant items quickly
  - Personalize user experience
  - Increase sales/revenue for business.
  - Increase user engagement
  - Help users discover new items

Basic Taxonomy: Taxonomy means classification of recommender systems into different types.

major types are:

1) Traditional / Non-personalized RS: Same recommendation is given to many/all users. ex. "Top 10 most popular movies".
It does not consider the individual user's preferences.

2) Content-Based RS: recommendations are based on characteristics/content of items and the user's previous preferences.
Ex: You watched avengers, Iron Man, Batman
The system identifies characteristics such as Action, Superhero, Adventure and recommends similar movies.

3) Collaborative filtering: recommendations are based on user-item ~~iterations~~ interactions such as ratings, likes, purchases, etc.
Example: If User A & User B have rated many movies similarly, movies liked by B but not yet watched by A can be recommended to A.

4) Knowledge-Based RS: based on explicit knowledge about the user's requirements and the items.
Ex: A car recommendation System asks:
budget = ₹ 10 lakh
Body = SUV
fuel = Petrol
→ rec cars satisfying these req's

Traditional & Non-Personalized: These are simplest rec systems.
- They generally recommend items based on overall popularity, rather than individual users.
ex: Most popular, Highest rated, Trending.

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Adv.

- Simple to implement
- Doesn't require much user info.
- Works even for new users.

Disadv.

- Same ~~recommend~~ rec may be shown to everyone
- Cannot understand individual preference
- Less personalized.

Overview of Data Mining methods for RS

- Data mining means extracting useful patterns and information from large amt of data.
- Rec systems collect data such as:
  - Ratings • Purchases • Searches • Clicks • Likes • Watch History
  - Browsing history • Reviews.
- This data is then analyzed to discover patterns.
- Imp. techniques:

A) Classification: Places data into predefined categories.
example: classifying a movie as: Action / Comedy / Horror / Romance.

B) Clustering: Group similar users or items together.
example: Users can be grouped as: Action-movie lovers, Comedy lovers.

C) Association Rule Mining: finds relationship b/w items.
example: Users who buy a laptop often buy a laptop bag.

D) Prediction: Predicts what a user may like or purchase.
example: User may give this movie a rating of 4.5/5.

Similarity Measures: determine how similar two users or two items are.

for example:

User	Movie A	Movie B	Movie C
user 1	5	4	1
user 2	5	4	2

Their ratings are similar, so the system may consider these users similar.

- Common similarity measures

A) Euclidean Distance: Measures the distance b/w two points.

$A(x_1, y_1)$ $B(x_2, y_2)$ $d = \sqrt{(x_1-x_2)^2 + (y_1-y_2)^2}$

Smaller distance → more similar.

B) Cosine Similarity: Measures the angle b/w two vectors.

$$\text{Cosine Similarity} = \frac{A \cdot B}{|A||B|}$$

The value generally ranges from: -1 to 1.

For RS applications, values closer to 1 indicate greater similarity.

C) Pearson Correlation: measures how strongly two variables are related.
- It is useful for comparing rating patterns b/w users.
- Value ranges from -1 to +1.
  - +1 : stronger positive relationship
  - 0 : no linear relationship
  - -1 : strong negative relationship.

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Dimensionality Reduction:

dimensionaly: A dataset may contain a very large number of features.
ex: a movie dataset could have:
Genre, Director, Actors, language, Year, Country, Ratings, Keywords, etc.
- This creates a high-dimensional dataset.
- Dimensionality Reduction means reducing the number of dimensions/features while retaining important information.
- we need it bc:
  - reduces computational cost
  - reduces storage requirements.
  - Removes unnecessary info.
  - Makes processing faster
  - Can improve Rec performance.

Singular Value Decomposition (SVD): is a mathematical technique used for dimensionality reduction & finding hidden patterns in data.
- A matrix A can be decomposed as: $A = U\Sigma V^T$
where
- U → represents information related to users
- Σ (Sigma): contains singular values/ imp. of latent features
- V^T: represents information related to items.

- SVD is a matrix factorization technique that decomposes a large matrix into smaller matrices and can be used to reduce dimensions and discover latent patterns in RS.

Application of RS RS is used in many domains.
1) E-commerce 2) Entertainment 3) Social Media
4) News 5) Education 6) Tourism.

Issues with RS: Cold-Start Problem: occurs when the system does not have enough information about a new user or new item to make good RS.
- There are mainly three types:

A) New User Cold Start: A new user joins the platform.
The system doesn't know: what does this user like?
ex: You create a new Netflix account.
Soln: The system may ask you to select some favourite genres/ movies.

B) New Item Cold Start: A new item is added to the system.
- The system doesn't have enough ~~item~~ interaction/rating data for it.
ex: A newly released movies has: 0 ratings, 0 view, 0 reviews.

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Sol^n: Use the item's content/features, or initially promote it to collect interaction data.

c) New System Cold Start: A completely new recommendation system has both new users and new items.

- There is almost no historical data available.

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RS Unit-2

Content Based Recommender System: recommends items that are similar to the items a user has liked in the past.

- Basic idea: User's past preferences → identify item features → find similar items → recommend them

ex: Suppose you watched: Interstellar, the Martian, Gravity
The system notices that these movies have features like sci-fi + space + Adventure. So it may recommend Arrival b/c its content/features a similar

- Content based rec system mainly uses information about the items, rather than comparing the user with other users.

Architecture:

Items → Extract Item features → Item Profiles → [matching System] → user Preferences
↓
recommendations ← Similarity / matching ← user profile

Main Components:

1) Item representation: • Describes an item using its features.
ex: A movie → genre, actor, director, language.

2) User Profile: • Represents what the user likes.
ex: User prefers → Scifi, english, space movies.

3) Matching function: Compares the user's profile with item profile

4) Recommendation: Items with high similarity are recommended.

Adv.

1) No need for other users: It can recommend items based on one users preference
2) Personalized: Rec. are specifically based on the user's interests.
3) Can rec. new items: If an item has sufficient feature information, it can be recommended even if nobody has rated it yet.
4) User preference is easy to understand
5) Works well for specialized interests.

Drawbacks

1) Limited rec./overspecialization: The system tends to recommend things very similar to what the user already likes.
- It may fail to introduce completely different but interesting items.
2) New-user problem: If the system doesn't know what a new user likes, it is difficult to create an accurate user profile.
3) feature extraction problem: Some items are difficult to describe using features.
ex: describing the "quality" or "emotion" of a song is difficult.

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4) Requires good item information : if item descriptions / features are poor, rec. will also be poor.

- **Content Representation**: means representing an item using a set of features that describe it.
  Ex: Movie : Interstella , a move can be represented as :
  → Genre : Sci-fi, Drama → language : English → Year : 2014.
  → Director : Christopher Nolan → Theme : Space

* Common Representations :
  • Keywords • Tags • Categories • Numerical features • Text descriptions
  • Metadata.

Content similarity: measures how similar two items are based on their features.
Ex: Movie A : Sci-fi, Space, Adventure
Movie B : Sci-fi, Space, Drama
They have several common features, So their similarity is high.

→ Common similarity measures: Cosine similarity is commonly used for text / features.
- another simple method is Jaccard similarity, which compares common features with total unique features.

Item Profiles : is a collection of features describing an item.
- Item profiles are needed b/c they allow the system to:
  Compare item profile → with user profile → Calculate similarity → Rec. items.

Discovering features of Data:
The system needs to discover useful features from item data.
Ex: "A scientist travels through space to find a new planet" movie description
system can identify features such as:
• Science, Space, Adventure, Planet.

Feature discovery can involve:
Keyword extraction, Text analysis, Meta data, Categories, Tags, NLP.

Obtaining Item features from Tags: Tags are labels associated with an item.
Ex: Interstellar Tags: Space, Science, future, astronomy, drama
The rec. uses these tags as features.

Adv:
Tags provide a relatively simple way to describe an item's content.

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Representing User Profiles: A user profile represents the user's interests and preferences.

- Suppose a user likes: Movie 1: sci-fi, space Movie 2: Sci-fi, future
  Movee 3: Space, Adventure.

The system can build a profile.
Sci-fi: High Space: High Adv.: Medium Future: Medium.

- The profile can be represented using weights.
  Higher weight = Stronger user interest.

Filtering: After creating the user profile, the system filters items acc. to how well they match the user's interest.

Similarity-Based Retrieval: This is the process of retrieving items that are most similar to the user's preferences.

Steps:
1) Create user profile 2) Represent each item using features
3) Calculate similarity b/w user profile & item profile.
4) Rank items
5) Recommend the top items.

Example:

Item	Similarity
Movie A	0.92
Movie B	0.81
Movie C	0.65
Movie D	0.30

The system reco Movie A & B.

Classification Algo: Classification can also be used to predict whether an item is suitable for a particular user.

- The system learns from the user's previous choices.

Ex:
- Decision Tree
- Naive Bayes
- K-Nearest Neighbors (KNN)
- Support Vector Machine (SVM)

Knowledge-Based Rec.

A Knowledge-Based Recommender system recommends items using explicit knowledge about the user, items, and domains.

- unlike content-based systems, it doesn't necessarily depend on user's previous ratings.

Ex: Laptop rec.
Budget = ₹ 70,000 Ram ≥ 16GB for gaming SSD required.

- The system uses it's knowledge of laptops and these req's to recommend suitable products.

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Knowledge Representation & Reasoning:

It means storing info. about:
Items, features, user req's, rel^{n}, constraints, Rules.

Reasoning: The system uses this stored knowledge to determine which items satisfy the user's req's.

User req's → Knowledge base → Reasoning → Suitable items.

Constraint-Based Recommender: recommends items that satisfy specific user constraints.

- The system eliminates products that doesn't satisfy these conditions.

Case-Based Recommender: looks at a user's current situation and finds similar previous cases/items.

ex: Suppose a user previously bought:
Phone: ₹ 30,000 good camera large battery. 5G.

Now, user wants another ~~good~~ phone with similar req's.

The system searches previous cases/products and finds similar ones.

Current problem → Find similar past case → Retrieve sol^{n} → Adapt sol^{n} if necessary → Reccamendation.